

Guided Capstone Project Report

Big Mountain Resort and Ticket Pricing Strategy

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Problem statement

At what price should Big Mountain Resort offer their lift tickets for the upcoming ski season, in order to counter-balance the \$~1.54 million increase in operating costs of the newly added ski-lift? [Find more details at this problem statement worksheet.](#)

Data Wrangling

The provided data was in the form of one (1) CSV file from the database manager, Alesha Eisen. After some initial investigation of the dataset's categorial and numerical features, we did a manual data correction step on a suspicious outlier value (ie Silverton Mountain, CO's skiable area or a resort that had been "open" for 2019 years) and features for which most of the data was missing/had a zero value (ie fastEight). We chose not to data correct other values based on the relevance of the entry with respect to our target feature of AdultWeekday (ticket price), including dropping rows without this feature.

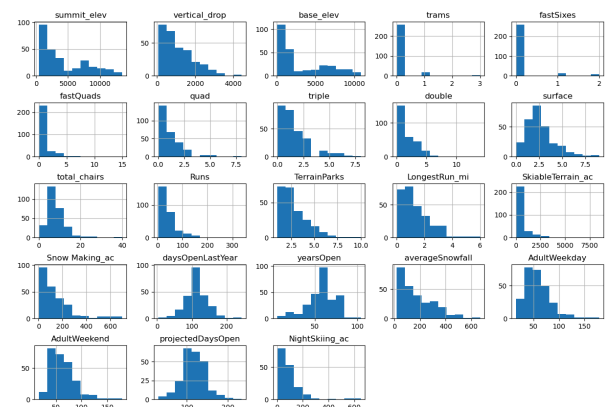


Figure 1: Distribution of the numerical data after our data correction step

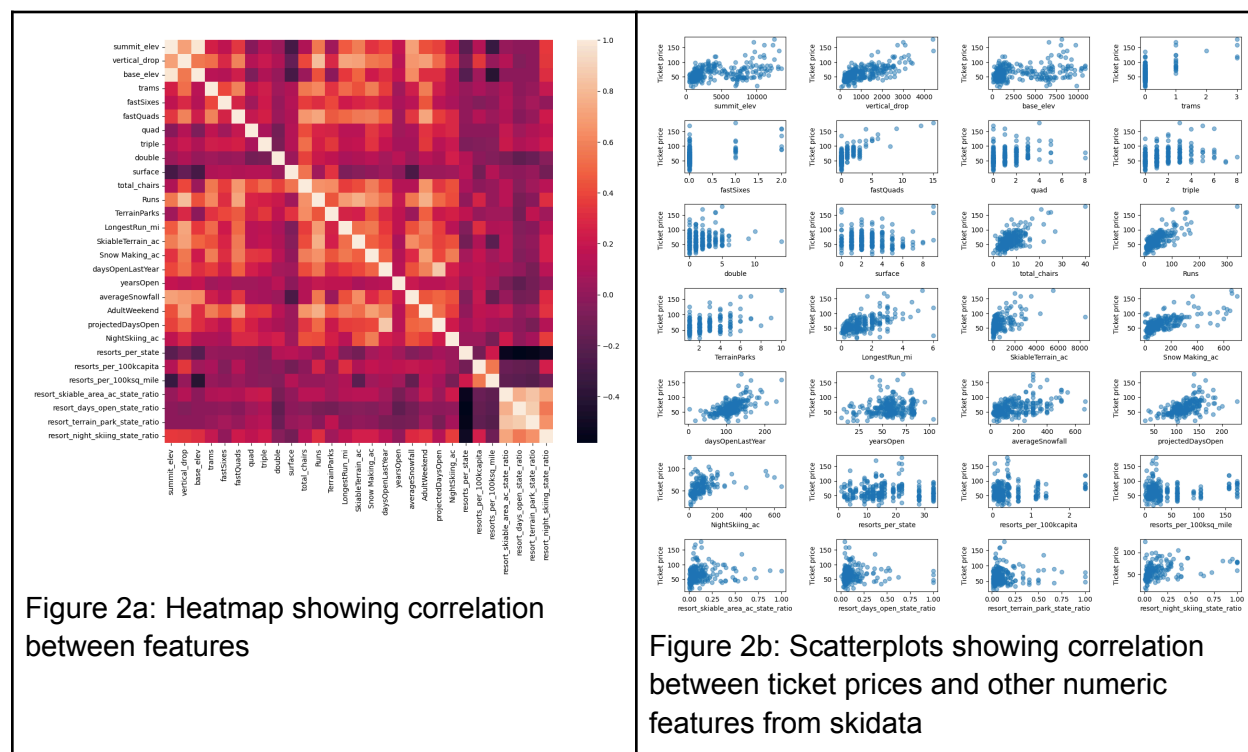
To complement these data, we also used information from [this Wikipedia page about US states](#) to include population information in the dataset. This required data cleaning resulting from how the web-scraping results were pulled into a pandas dataframe.

Finally, we chose for our target feature the ticket prices of AdultWeekday (instead of AdultWeekend). The resorts in Montana showed no difference between the two columns, and the AdultWeekday column had fewer missing values.

Exploratory Data Analysis

State population data was incorporated into our existing ski data set during our first pass at analysis. We also undertook principle component analysis (PCA) to determine if there was much state-to-state variance. We found there was not, and could treat the state data as a set (e.g., without special considerations for subsets of this dataset)

Then, we did some preliminary correlation on our numeric features (see figure 2a) and between ticket price and the numeric features (figure 2b)



Model Preprocessing with feature engineering / Algorithms used to build the model with evaluation metric

We split our data for a 70/30 train/test split. Features with missing values were set to the median. After further model refinement and cross-validation, the pipeline chosen was (1) SimpleImputer with median value (2) StandardScaler and (3) RandomForestRegressor. The random forest model showed less variability and has a lower cross-validation MAE when compared to the linear regression model. We verified that our data was sufficient to use this model.

Winning model and scenario modelling

Model chosen: RandomForest regression

Modeled price: \$95.87 with mean absolute error of \$10.39

Pricing recommendation

Big Mountain Resort currently charges \$81 for an adult lift ticket (both on weekdays and weekends). Based on the results of our modeling, we find that a ticket price of \$95.87 (with MAE of \$10.39 which could further push ticket price up to \$106.26) can be supported in the marketplace based on the facilities the resort has.

Based on available data, BMR is already the highest priced resort in Montana. An increase in ticket price could cement its reputation as the premier resort in Big Sky Country.

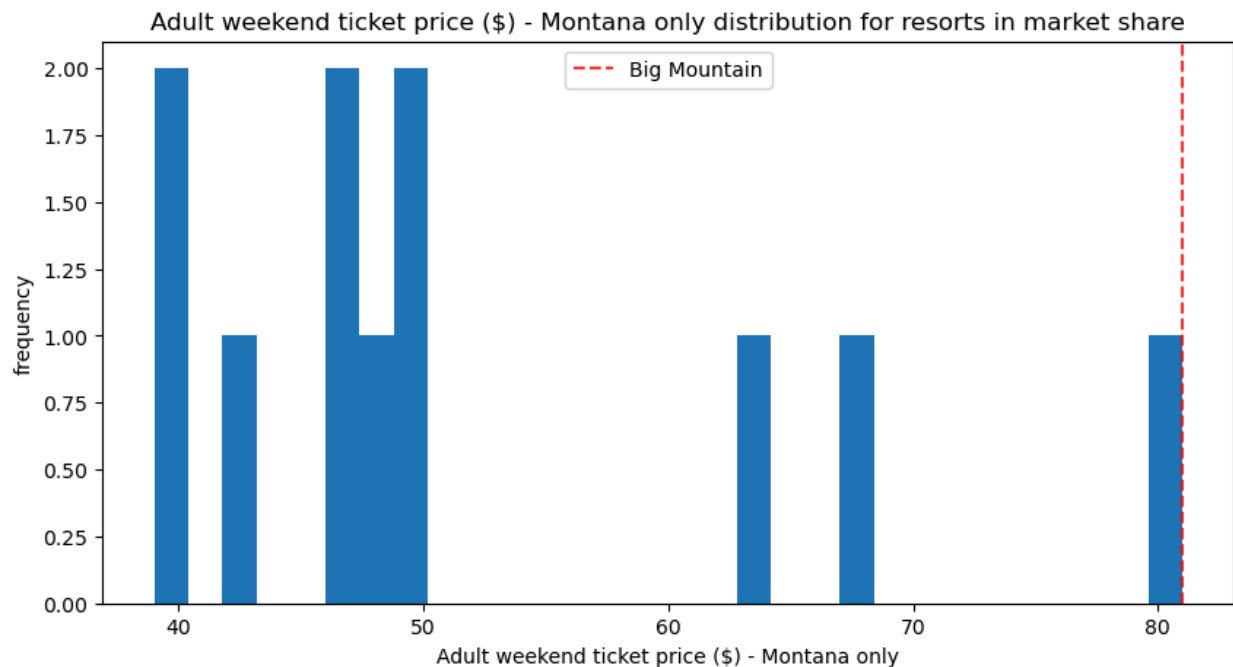


Figure 3: Ticket pricing of adult weekend tickets for resorts in Montana.

Conclusion

The scope of our analyses suggests that the facilities at BMR support an increase in ticket prices. Given this increase, the new chairlift could be paid for with income from the next ski season.

Preliminary analyses also suggest that the features of vertical drop, snow making acres, total ski lifts, number of fast quad lifts, amount of runs, longest ski run, number of trams, and skiable terrain have a large effect on ticket prices. To that extent, in the exploration of the four (4)

scenarios posited in this process, the most viable suggestions are to shut down up to ~5 ski runs, or to add more vertical drop by building another new chairlift (supports an increase in ticket price by another \$~2).

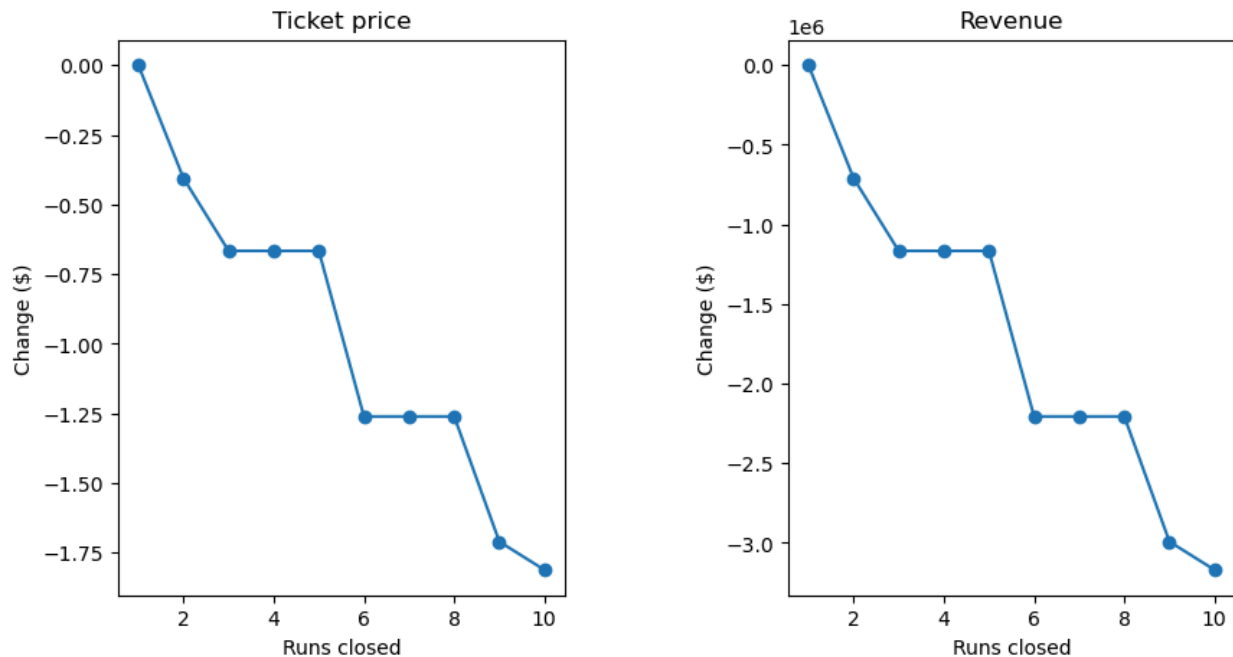


Figure 4: Analyses of ticket price and revenue from our model for closing up to 10 ski runs.

Future scope of work

First, we are aware that the data set is incomplete because of the missing values. Further, we know from the row about Silverton Mountain in CO that it is not guaranteed that every entry is correct. Improvements to the dataset can help our model with a more accurate depiction of reality.

Further, this work only takes into account the information about the skiing portion. While Big Mountain Resort's initial attraction is its ski facilities, this model does not account for other aspects of the resort. For example, perhaps a lower ticket price is subsidized by inflated food or beverage costs at on-site restaurants. Information about places to eat, lodging, or other resort services like spas or shopping could give a fuller picture of BMR's facilities – not just those related to skiing.

Also, information about operating costs could be a nice feature to include in the future if our goal is to maximize revenue and not just income.